IMED BEN SLIMENE

Ph.D. in Applied Geology | Hydrogeologist & GIS/Remote Sensing Specialist

Nabeul 8090, Tunisia | +216 25 167 447 | imedlcd@gmail.com

ORCID: https://orcid.org/0009-0007-6490-530X **Researchgate:** **https://www.researchgate.net/profile/Imed-Ben-Slimene**

PROFESSIONAL SUMMARY

Researcher and educator with a Ph.D. in Applied Geology (University of Carthage, 2025, Honours with Jury Commendation) and over a decade of combined academic and field experience across Tunisia, Saudi Arabia, and Germany. Core expertise spans groundwater hydrogeochemistry, structural hydrogeology, GIS/remote sensing, and machine learning applied to water-resource management. Author of 5 peer-reviewed publications (Q1/Q2 journals Water, J. African Earth Sciences, Soil & Sediment Contamination, Acque Sotterranee) and 1 accepted manuscript; 2 additional papers under review. International research mobility includes two internships at UFZ-Helmholtz Centre (Halle, Germany) and invited seminars at UQAM (Montréal). Proven track record in multi-institutional collaboration, graduate supervision, and science communication in English, French, and Arabic.

RESEARCH INTERESTS

Groundwater modelling, Hydrogeochemistry, GIS & Remote Sensing, Machine Learning for Water Resources

EDUCATION

Ph.D. in Earth Sciences & Technologies - Applied Geology

Faculty of Sciences of Bizerte, University of Carthage, Tunisia | 2022–2025

Mention: Highly Commended with Honors from the Jury (Highest Distinction)

- Thesis: Impact of the structure of the foreland basins associated with the Téboursouk overlap on the hydrodynamic and hydrochemical behavior of the associated aquifer systems.
- Integrated lineament extraction (remote sensing + KDE), machine learning (Random Forest, KNN, ANN — $R^2 > 0.90$), stable isotope analysis ($\delta^{18}O$, $\delta^{2}H$, $\delta^{13}C$) and multivariate statistics (PCA, HCA).
- Analytical collaboration with UFZ–Helmholtz Centre, Halle, Germany (Dr S. Geyer, Dr C. Müller). DOI: 10.13140/RG.2.2.34899.03361/1

M.Sc. in Desertification Control

National Agronomic Institute of Tunisia (INAT) | 2007–2009

- Estimation of natural recharge of Plio-Quaternary aquifer at Mouares-Redeyef and its impact on water renewal, using the WetSpass model coupled with spatialised hydroclimatic GIS data.
- Results presented at the Tuniso-Algerian Colloquium (2008) and published in IJCIET (2012). DOI: 10.13140/RG.2.2.24709.05603

B.Sc. in Earth Sciences-Geology: Mapping and Planning

Faculty of Mathematics, Physics, and Natural Sciences, Tunis (FST) | 2001–2005

- Title of Final Project: Representations of the Earth from Space
- From Antiquity to Greek Science: The Transition from a Symbolic View to the Concept of the Earth's Sphericity.
- Renaissance: The Rise of Projections and Improvements in Cartographic Accuracy.
- Modern Era: The Digital Revolution (GIS, GNSS, remote sensing).
- Today: Space-based mapping (satellites, orthophotos), precise and multiscale

RESEARCH EXPERIENCE

Research Intern - Catchment Hydrology Department

UFZ - Helmholtz Centre for Environmental Research, Halle, Germany

***Internship 1**: Sep–Nov 2023 | **Internship 2**: Jul–Oct 2024*

- Produced reference analytical datasets for major ions, trace elements, and stable isotopes from 50 Tunisian groundwater samples used directly in 3 published articles.
- Acquired operational competence in numerical groundwater flow and solute-transport modelling (MODFLOW, MT3D) under UFZ protocols.
- Attended specialised training at Martin Luther University Halle-Wittenberg: Numerical Groundwater Flow & Transport Modelling, 3D Hydrogeology & Environmental Geology (open-source GIS), Python Programming with GIS.
- Presented research at the UFZ Isotope Workshop Science Days and the 8th UFZ Stable Isotope Platform Meeting (Nov 2023).

Researcher - ELMAA Project (Contract No. 015410)

National Agronomic Institute of Tunisia (INAT) | EU Programme PF6-2003-INCO-MPC-2 | Nov 2007 – Mar 2009

- Contributed to hydrological and hydrogeological investigations of the Gafsa mining basin.
- Developed the project GIS, integrating multi-source spatial data for resource mapping and decision support.

TEACHING & ACADEMIC EXPERIENCE

Instructor in Topography & Geomatics (Civil Engineering & Architecture)

Tabuk College of Technology (TVTC), Kingdom of Saudi Arabia | Aug 2013 – Aug 2020

- Delivered courses in Topography I–IV, Computer Topography I–II, GPS Systems, Surveying, Topographic Calculations, Instrument Calibration, GIS, Photogrammetry, Remote Sensing and Surveyor Management to Technical Diploma students.
- Served as Academic Planning Officer for students and faculty (Aug 2015 – Jun 2020) and Equipment Manager (Aug 2019 – Sep 2020).
- Managed quality assurance of teaching materials and adapted curricula to TVTC competency frameworks.

Sessional Lecturer - Earth Sciences & Geomatics

Higher Institute of Environmental Sciences & Technologies (ISSTE), Borj Cédria, Tunisia | 2022–2023

- Taught Hydrogeological Modelling & Simulation, GIS, Remote Sensing, Structural Geology, and Geomatics to undergraduate and postgraduate students.
- Designed competency-aligned teaching materials matched to B.Sc. and M.Sc. programme specifications.

Sessional Lecturer - Geology & Computer Science

Faculty of Sciences of Gafsa | 2011–2013

- Courses included: GIS, Remote Sensing, Structural Geology, Petrology & Petrogenesis, Geology & Reservoirs of Tunisia, Fracture Reservoir Modelling, Algorithmic & C Programming, Object-Oriented Programming.

Graduate Thesis Co-supervisor

ISSTE Borj Cédria | 2025

- Co-supervised M.Sc. dissertation: "Contribution of Hydrogeochemical Modelling to the Study of Groundwater Contamination Risks in the El Haouaria, Cap Bon Plain" (defended Apr 2025).

PEER-REVIEWED PUBLICATIONS

- **[Published]** **Ben Slimene, I**. et al. (2025). Geospatial Mapping and Multivariate Statistical Analysis for Assessing Groundwater Quality in Bou Arada–El Aroussa Plain. ***Acque Sotterranee - Italian Journal of Groundwater***. https://doi.org/10.7343/as-2025-816

- **[Published]** **Ben Slimene, I**. et al. (2026). Integrating Remote Sensing, GIS, and Machine Learning for Lineament Analysis and Hydrogeological Assessment in the Northern Atlassic Domain of Tunisia. ***Journal of African Earth Sciences.*** https://doi.org/10.1016/j.jafrearsci.2026.106210
- **[Published]** **Ben Slimene, I**. et al. Predictive Modeling and Spatial Mapping of Groundwater Quality for Irrigation Using Integrated Indices and Machine Learning: Application to the Sidi Bou Rouis–El Krib–Borj El Messaoudi Region. ***Euro-Mediterranean Journal for Environmental Integration***. https://doi.org/10.1007/s41207-026-01202-z
- **[Published]** Sarra Bel Haj Salem, Aissam Gaagai ,**Imed Ben Slimene** et al. (2023). Applying Multivariate Analysis and Machine Learning Approaches to Evaluating Groundwater Quality on the Kairouan Plain, Tunisia. ***Water (MDPI).*** https://doi.org/10.3390/w15193495
- **[Published]** Sonia Dhaouadi, Samir Ghannem, Sabri Kanzari, **Imed Ben Slimene** et al. (2023). Assessment of Heavy Metals Contamination and Potential Ecological Risk in Surface Sediments of the El Bey Wadi, Tunisia. ***Soil and Sediment Contamination.*** https://doi.org/10.1080/15320383.2023.2271987
- **[Published]** Nadia Khelif; **Imed Ben Slimène**; M.Moncef Chalbaoui. (2012). Intrinsic Vulnerability Analysis to Nitrate Contamination. International Journal of Civil Engineering and Technology. ***International Journal of Civil Engineering and Technology (IJCIET),*** ISSN 0976 – 6308 (Print), ISSN 0976 – 6316(Online) Volume 3, Issue 2, July- December (2012), © IAEME
- **[Under review]** Eya HEDI, Amor BEN MOUSSA, **Imed BEN SLIMENE**, Stefan GEYER. Hydrogeochemical Assessment and Groundwater Quality Modelling Using Machine Learning Approaches in the El Haouaria Plain, NE Tunisia. (Submitted).

CONFERENCE PRESENTATIONS & SCIENTIFIC EVENTS

- Ben Slimene, I. et al. (2025). Remote Sensing, GIS, and ML for Lineament Analysis and Hydrogeological Assessment of the Bou Arada–El Aroussa Plain. International Days of Geology & Geotechnics (J2G), 9–10 May 2025.
- Ben Slimene, I. (2024). Groundwater Vulnerability Mapping. Regular Seminar, GEOTOP – Université du Québec à Montréal (UQAM), 16 Jan 2024. [Invited]
- Ben Slimene, I. (2024). Impact of Téboursouk Thrust Basins on Aquifer Hydrodynamics and Hydrochemistry. Regular Seminar, GEOTOP – UQAM, 1 Feb 2024. [Invited]
- Ben Slimene, I. (2023). The Isotope Workshop UFZ Science Days — Presentation. UFZ–Helmholtz Centre for Environmental Research, Germany, 3 Nov 2023.
- Ben Slimene, I. (2023). 8th UFZ Stable Isotope Platform Meeting — Participation. UFZ–Helmholtz Centre for Environmental Research, Germany, 17 Nov 2023.
- Ben Slimene, I. (2023). Assessment of Emerging Organic Pollution, Ecology & Modelling of Mediterranean Hydrosystems. INWAT Project Workshop, Hammamet, Tunisia, 27 Mar 2023.
- Ben Slimene, I. (2022). Emerging Contaminants in Groundwater: Environments & Human Health. INWAT Project Stakeholders Workshop, Hammamet, Tunisia, 5 Apr 2022.
- Ben Slimene, I. (2009). Evaluation of Aquifer Renewal Rates in the Gafsa Mining Basin under Extreme Climatic Trends. CPG Conference, 23 Apr 2009. [Poster]
- Ben Slimene, I. (2008). Multi-year Recharge Estimation of the Plio-Quaternary Aquifer of Redeyef–Moularès (Gafsa). 1st Tuniso-Algerian University & Researchers Colloquium, 27–28 Mar 2008.

CORE COMPETENCIES & TECHNICAL SKILLS

GIS & Remote Sensing

- ArcGIS (v10.2+), QGIS, IDRISI, spatial analysis, thematic mapping, multi-criteria AHP
- ENVI 4.1 and PCI Geomatica 2018: satellite image processing, lineament extraction, land-use classification
- KDE analysis, automated lineament detection, terrain analysis (DEM, slope, aspect)

Hydrogeological & Hydrological Modelling

- MODFLOW / MT3D (groundwater flow & solute transport), GMS 5.0
- WetSpass (water balance, recharge estimation), WEAP (integrated water planning)

- RockWorks 17 geological, hydrogeological, and hydrochemical 3D modelling

Hydrochemical & Isotopic Analysis

- Piper, Gibbs, Chadha, Schoeller diagrams; SAR, IWQI, RSC, Na% irrigation indices
- Stable isotope interpretation ($\delta^{18}O$, $\delta^{2}H$, $\delta^{13}C$)
- EPA PMF 5.0 source apportionment; multivariate statistics (PCA, HCA, ANOVA)

Programming & Data Science

- Python (SciPy, Pandas, Matplotlib, Scikit-learn) applied to geosciences and GIS
- Machine learning: Random Forest, KNN, ANN for water-quality classification and prediction
- LaTeX for scientific writing; SQL for spatial database management; C / OOP

Scientific Communication & Management

- Manuscript preparation for Q1/Q2 Scopus/WoS journals; peer-review experience
- Grant writing, project reporting (EU ELMAA, INWAT)
- Academic planning, curriculum design, graduate supervision

SELECTED TRAINING & CERTIFICATIONS

- Numerical Groundwater Flow Modelling - Martin Luther University Halle-Wittenberg (Nov 2023)
- Numerical Groundwater Transport Modelling - Martin Luther University Halle-Wittenberg (Sep-Oct 2023)
- 3D Hydrogeology & Environmental Geology (Open-Source GIS) - Martin Luther University (Oct 2023)
- Python Programming with GIS - Martin Luther University Halle-Wittenberg (Nov 2023)
- LaTeX, Python & Scientific Computing - Faculty of Sciences of Bizerte (Feb 2024)
- Scientific Research Methodology - CNUDST, Tunisia (May 2023)
- GMS Hydrogeological Modelling - Tunisia (Nov 2008)
- WEAP Water Resource Planning Software - Tunisia (Jan 2009)
- Civil 3D (AutoCAD) - Tabuk College of Technology, KSA (Sep 2016)
- GPS ProMark 120 - Tabuk College of Technology, KSA (Mar 2014)

LANGUAGES

Arabic: Native | French: Proficient (C1) | English: Intermediate (B2, actively improving)

REFERENCES

Mc. Amor Ben Moussa
Maître de conférences
Higher Institute of Environmental Sciences and Technologies (ISSTE)
University of Carthage, Tunisia
✉ omar.benmousa@isste.u-carthage.tn
✉ amor_geologie@yahoo.fr
Phone: +216 54 249 246

Dr. Stefan Geyer
Groundwater Resources Research Group
Department of Catchment Hydrology
Helmholtz Centre for Environmental Research (UFZ), Halle, Germany
✉ stefan.geyer@ufz.de
Phone: +49 341 6025 4217